

class14

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Table of contents

Differential Expression Analysis	1
Run DESeq2	5
Get DESeq2 results	7
Volcano Plot	8
Add gene annotation	11
Section 2: Pathway Analysis	14
Create named fold change vector for GAGE	15
Run KEGG pathway analysis	16
Pathview example: Cell cycle pathway	17
Top 5 upregulated KEGG pathways	18
Section 3: Gene Ontology Analysis	20
Section 4: Reactome Analysis	22

Differential Expression Analysis

```
library(DESeq2)
```

Loading required package: S4Vectors

Loading required package: stats4

Loading required package: BiocGenerics

Loading required package: generics

Attaching package: 'generics'

The following objects are masked from 'package:base':

as.difftime, as.factor, as.ordered, intersect, is.element, setdiff,
setequal, union

Attaching package: 'BiocGenerics'

The following objects are masked from 'package:stats':

IQR, mad, sd, var, xtabs

The following objects are masked from 'package:base':

anyDuplicated, aperm, append, as.data.frame, basename, cbind,
colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,
get, grep, grepl, is.unsorted, lapply, Map, mapply, match, mget,
order, paste, pmax, pmax.int, pmin, pmin.int, Position, rank,
rbind, Reduce, rownames, sapply, saveRDS, table, tapply, unique,
unsplit, which.max, which.min

Attaching package: 'S4Vectors'

The following object is masked from 'package:utils':

findMatches

The following objects are masked from 'package:base':

expand.grid, I, unname

Loading required package: IRanges

Loading required package: GenomicRanges

Loading required package: Seqinfo

Loading required package: SummarizedExperiment

Loading required package: MatrixGenerics

Loading required package: matrixStats

Attaching package: 'MatrixGenerics'

The following objects are masked from 'package:matrixStats':

colAlls, colAnyNAs, colAnys, colAvgPerRowSet, colCollapse,
colCounts, colCummaxs, colCummins, colCumprods, colCumsums,
colDiffs, colIQRDiffs, colIQRs, colLogSumExps, colMadDiffs,
colMads, colMaxs, colMeans2, colMedians, colMins, colOrderStats,
colProds, colQuantiles, colRanges, colRanks, colSdDiffs, colSds,
colSums2, colTabulates, colVarDiffs, colVars, colWeightedMads,
colWeightedMeans, colWeightedMedians, colWeightedSds,
colWeightedVars, rowAlls, rowAnyNAs, rowAnys, rowAvgPerColSet,
rowCollapse, rowCounts, rowCummaxs, rowCummins, rowCumprods,
rowCumsums, rowDiffs, rowIQRDiffs, rowIQRs, rowLogSumExps,
rowMadDiffs, rowMads, rowMaxs, rowMeans2, rowMedians, rowMins,
rowOrderStats, rowProds, rowQuantiles, rowRanges, rowRanks,
rowSdDiffs, rowSds, rowSums2, rowTabulates, rowVarDiffs, rowVars,
rowWeightedMads, rowWeightedMeans, rowWeightedMedians,
rowWeightedSds, rowWeightedVars

Loading required package: Biobase

Welcome to Bioconductor

Vignettes contain introductory material; view with
'browseVignettes()'. To cite Bioconductor, see
'citation("Biobase")', and for packages 'citation("pkgname")'.

Attaching package: 'Biobase'

The following object is masked from 'package:MatrixGenerics':

rowMedians

The following objects are masked from 'package:matrixStats':

anyMissing, rowMedians

```
metaFile <- "GSE37704_metadata.csv"
countFile <- "GSE37704_featurecounts.csv"

# Import metadata
colData <- read.csv(metaFile, row.names = 1)
head(colData)
```

```
              condition
SRR493366 control_sirna
SRR493367 control_sirna
SRR493368 control_sirna
SRR493369      hoxa1_kd
SRR493370      hoxa1_kd
SRR493371      hoxa1_kd
```

```
# Import count data
countData <- read.csv(countFile, row.names = 1)
head(countData)
```

	length	SRR493366	SRR493367	SRR493368	SRR493369	SRR493370
ENSG00000186092	918	0	0	0	0	0
ENSG00000279928	718	0	0	0	0	0
ENSG00000279457	1982	23	28	29	29	28
ENSG00000278566	939	0	0	0	0	0
ENSG00000273547	939	0	0	0	0	0
ENSG00000187634	3214	124	123	205	207	212
		SRR493371				
ENSG00000186092		0				
ENSG00000279928		0				
ENSG00000279457		46				

ENSG00000278566	0
ENSG00000273547	0
ENSG00000187634	258

Q1. Remove the troublesome first column from countData

```
# The first column is "length", which is not a sample count column
# We remove column 1 and keep only the sample count columns
countData <- as.matrix(countData[, -1])

# Check that the length column is gone
head(countData)
```

	SRR493366	SRR493367	SRR493368	SRR493369	SRR493370	SRR493371
ENSG00000186092	0	0	0	0	0	0
ENSG00000279928	0	0	0	0	0	0
ENSG00000279457	23	28	29	29	28	46
ENSG00000278566	0	0	0	0	0	0
ENSG00000273547	0	0	0	0	0	0
ENSG00000187634	124	123	205	207	212	258

Q2. Filter out genes with zero reads across all samples

```
# rowSums(countData) adds the counts across all samples for each gene
# We keep only genes where the total count is greater than 0
countData <- countData[rowSums(countData) > 0, ]

# Check filtered count data
head(countData)
```

	SRR493366	SRR493367	SRR493368	SRR493369	SRR493370	SRR493371
ENSG00000279457	23	28	29	29	28	46
ENSG00000187634	124	123	205	207	212	258
ENSG00000188976	1637	1831	2383	1226	1326	1504
ENSG00000187961	120	153	180	236	255	357
ENSG00000187583	24	48	65	44	48	64
ENSG00000187642	4	9	16	14	16	16

Run DESeq2

```
# Create DESeq2 dataset object
# countData is the gene count matrix
# colData is the metadata
# design is the condition being tested
dds <- DESeqDataSetFromMatrix(
  countData = countData,
  colData = colData,
  design = ~ condition
)
```

Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in design formula are characters, converting to factors

```
# Run DESeq2 pipeline
dds <- DESeq(dds)
```

estimating size factors

estimating dispersions

gene-wise dispersion estimates

mean-dispersion relationship

final dispersion estimates

fitting model and testing

```
dds
```

```
class: DESeqDataSet
dim: 15975 6
metadata(1): version
assays(4): counts mu H cooks
rownames(15975): ENSG00000279457 ENSG00000187634 ... ENSG00000276345
               ENSG00000271254
rowData names(22): baseMean baseVar ... deviance maxCooks
colnames(6): SRR493366 SRR493367 ... SRR493370 SRR493371
colData names(2): condition sizeFactor
```

Get DESeq2 results

```
# Get differential expression results
res <- results(dds)

# Check result names
resultsNames(dds)
```

```
[1] "Intercept" "condition_hoxa1_kd_vs_control_sirna"
```

```
# Preview results
head(res)
```

```
log2 fold change (MLE): condition hoxa1 kd vs control sirna
Wald test p-value: condition hoxa1 kd vs control sirna
DataFrame with 6 rows and 6 columns
```

	baseMean	log2FoldChange	lfcSE	stat	pvalue
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
ENSG00000279457	29.9136	0.1792571	0.3248215	0.551863	5.81042e-01
ENSG00000187634	183.2296	0.4264571	0.1402658	3.040350	2.36304e-03
ENSG00000188976	1651.1881	-0.6927205	0.0548465	-12.630156	1.43993e-36
ENSG00000187961	209.6379	0.7297556	0.1318599	5.534326	3.12428e-08
ENSG00000187583	47.2551	0.0405765	0.2718928	0.149237	8.81366e-01
ENSG00000187642	11.9798	0.5428105	0.5215598	1.040744	2.97994e-01

```
      padj
      <numeric>
ENSG00000279457 6.86555e-01
ENSG00000187634 5.15718e-03
ENSG00000188976 1.76553e-35
ENSG00000187961 1.13413e-07
ENSG00000187583 9.19031e-01
ENSG00000187642 4.03379e-01
```

Q3. Summarize DESeq2 results

```
# Summarize number of upregulated and downregulated genes
summary(res)
```

```
out of 15975 with nonzero total read count
```

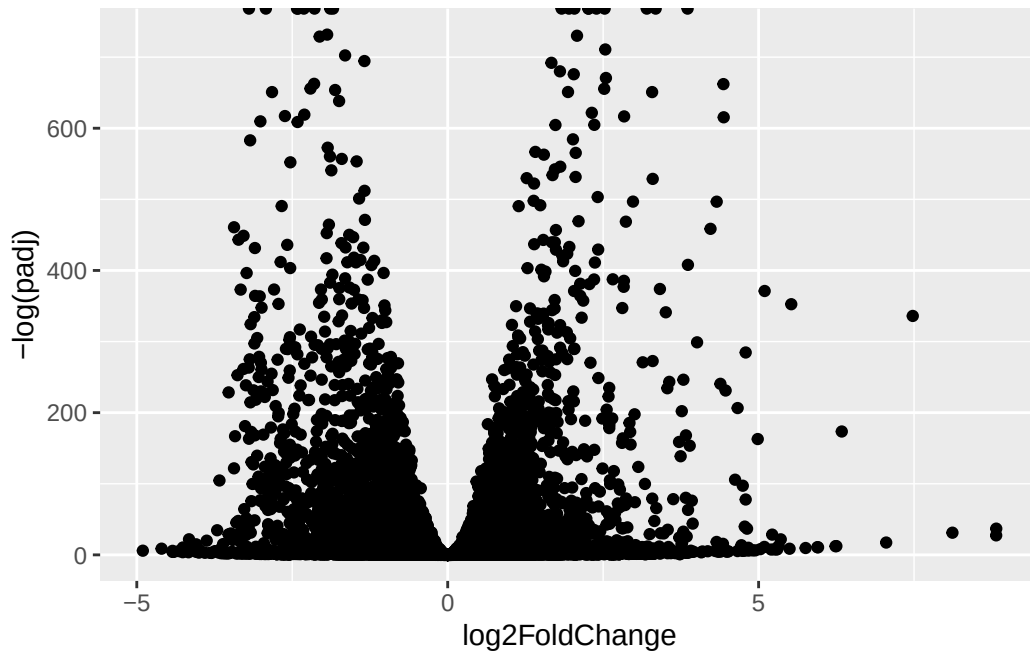
```
adjusted p-value < 0.1
LFC > 0 (up)      : 4349, 27%
LFC < 0 (down)    : 4396, 28%
outliers [1]      : 0, 0%
low counts [2]    : 1237, 7.7%
(mean count < 0)
[1] see 'cooksCutoff' argument of ?results
[2] see 'independentFiltering' argument of ?results
```

Volcano Plot

```
# Load ggplot2 for plotting
library(ggplot2)

# Basic volcano plot:
# x-axis = log2 fold change
# y-axis = -log adjusted p-value
ggplot(as.data.frame(res)) +
  aes(
    x = log2FoldChange,
    y = -log(padj)
  ) +
  geom_point()
```

Warning: Removed 1237 rows containing missing values or values outside the scale range (``geom_point()``).



Q4. Improved volcano plot with colors and cutoff lines

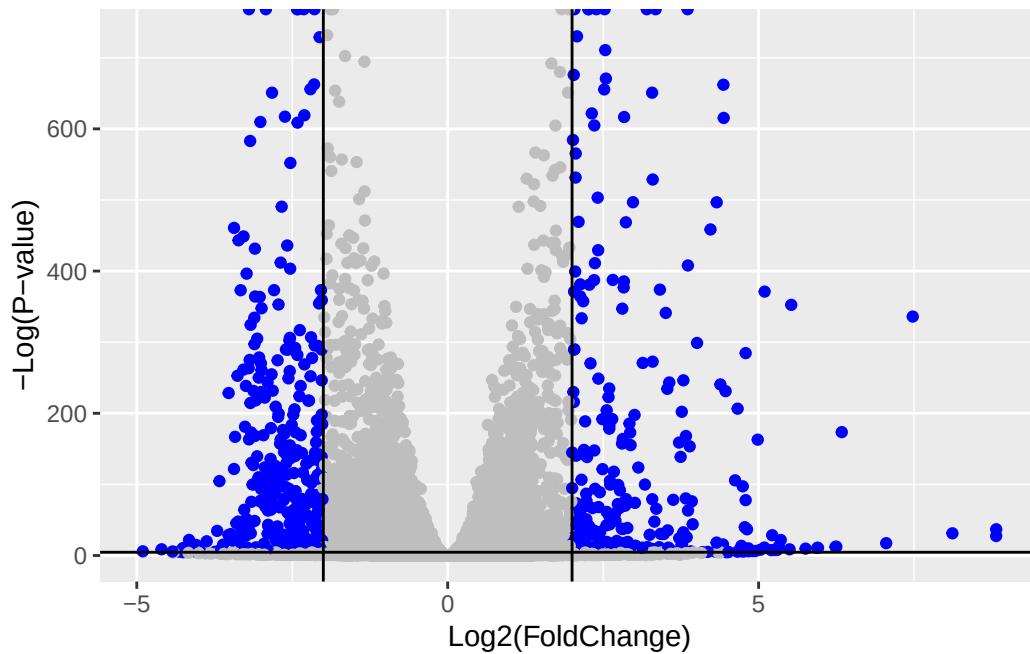
```
# Make a color vector for all genes
mycols <- rep("gray", nrow(res))

# Color blue genes with fold change above 2
mycols[abs(res$log2FoldChange) > 2] <- "blue"

# Color gray genes with adjusted p-value greater than 0.01
mycols[res$padj > 0.01] <- "gray"

# Make volcano plot
ggplot(as.data.frame(res)) +
  aes(
    x = log2FoldChange,
    y = -log(padj)
  ) +
  geom_point(color = mycols) +
  xlab("Log2(FoldChange)") +
  ylab("-Log(P-value)") +
  geom_vline(xintercept = c(-2, 2)) +
  geom_hline(yintercept = -log(0.01))
```

Warning: Removed 1237 rows containing missing values or values outside the scale range (`geom\_point()`).



```
# Create logical vector for significant genes
sig <- res$padj < 0.05 &
      abs(res$log2FoldChange) > 2

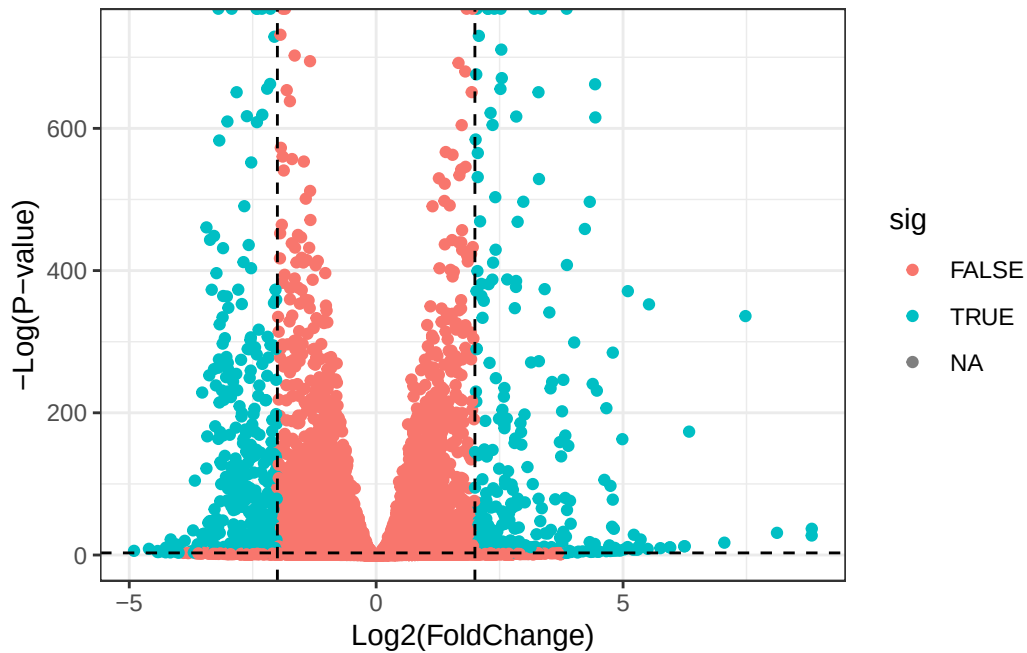
# Make volcano plot highlighting significant genes
ggplot(as.data.frame(res)) +
  aes(
    x = log2FoldChange,
    y = -log(padj),
    color = sig
  ) +
  geom_point() +
  xlab("Log2(FoldChange)") +
  ylab("-Log(P-value)") +
  geom_vline(
    xintercept = c(-2, 2),
    linetype = "dashed"
  ) +
  geom_hline(
```

```

yintercept = -log(0.05),
linetype = "dashed"
) +
theme_bw()

```

Warning: Removed 1237 rows containing missing values or values outside the scale range (`geom\_point()`).



Add gene annotation

```

# Load annotation packages
library(AnnotationDbi)
library(org.Hs.eg.db)

```

```

# Show available annotation columns
columns(org.Hs.eg.db)

```

[1]	"ACCNUM"	"ALIAS"	"ENSEMBL"	"ENSEMBLPROT"	"ENSEMBLTRANS"
[6]	"ENTREZID"	"ENZYME"	"EVIDENCE"	"EVIDENCEALL"	"GENENAME"
[11]	"GENETYPE"	"GO"	"GOALL"	"IPI"	"MAP"
[16]	"OMIM"	"ONTOLOGY"	"ONTOLOGYALL"	"PATH"	"PFAM"
[21]	"PMID"	"PROSITE"	"REFSEQ"	"SYMBOL"	"UCSCKG"
[26]	"UNIPROT"				

Q5. Add SYMBOL, ENTREZID, and GENENAME annotations

```
# Add gene symbols
res$symbol <- mapIds(
  org.Hs.eg.db,
  keys = row.names(res),
  keytype = "ENSEMBL",
  column = "SYMBOL",
  multiVals = "first"
)
```

'select()' returned 1:many mapping between keys and columns

```
# Add Entrez IDs
res$entrez <- mapIds(
  org.Hs.eg.db,
  keys = row.names(res),
  keytype = "ENSEMBL",
  column = "ENTREZID",
  multiVals = "first"
)
```

'select()' returned 1:many mapping between keys and columns

```
# Add full gene names
res$name <- mapIds(
  org.Hs.eg.db,
  keys = row.names(res),
  keytype = "ENSEMBL",
  column = "GENENAME",
  multiVals = "first"
)
```

'select()' returned 1:many mapping between keys and columns

```
# Preview annotated results
head(res, 10)
```

log2 fold change (MLE): condition hoxa1 kd vs control sirna

Wald test p-value: condition hoxa1 kd vs control sirna

DataFrame with 10 rows and 9 columns

	baseMean	log2FoldChange	lfcSE	stat	pvalue
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
ENSG00000279457	29.913579	0.1792571	0.3248215	0.551863	5.81042e-01
ENSG00000187634	183.229650	0.4264571	0.1402658	3.040350	2.36304e-03
ENSG00000188976	1651.188076	-0.6927205	0.0548465	-12.630156	1.43993e-36
ENSG00000187961	209.637938	0.7297556	0.1318599	5.534326	3.12428e-08
ENSG00000187583	47.255123	0.0405765	0.2718928	0.149237	8.81366e-01
ENSG00000187642	11.979750	0.5428105	0.5215598	1.040744	2.97994e-01
ENSG00000188290	108.922128	2.0570638	0.1969053	10.446970	1.51281e-25
ENSG00000187608	350.716868	0.2573837	0.1027266	2.505522	1.22271e-02
ENSG00000188157	9128.439422	0.3899088	0.0467164	8.346302	7.04333e-17
ENSG00000237330	0.158192	0.7859552	4.0804729	0.192614	8.47261e-01
	padj	symbol	entrez	name	
	<numeric>	<character>	<character>	<character>	
ENSG00000279457	6.86555e-01	NA	NA	NA	
ENSG00000187634	5.15718e-03	SAMD11	148398	sterile alpha motif ..	
ENSG00000188976	1.76553e-35	NOC2L	26155	NOC2 like nucleolar ..	
ENSG00000187961	1.13413e-07	KLHL17	339451	kelch like family me..	
ENSG00000187583	9.19031e-01	PLEKHN1	84069	pleckstrin homology ..	
ENSG00000187642	4.03379e-01	PERM1	84808	PPARGC1 and ESRR ind..	
ENSG00000188290	1.30538e-24	HES4	57801	hes family bHLH tran..	
ENSG00000187608	2.37452e-02	ISG15	9636	ISG15 ubiquitin like..	
ENSG00000188157	4.21970e-16	AGRN	375790	agrin	
ENSG00000237330	NA	RNF223	401934	ring finger protein ..	

Q6. Reorder results and save to CSV

```
# Order results by adjusted p-value
res <- res[order(res$padj), ]

# Save results to CSV file
write.csv(res, file = "deseq_results.csv")
```

Section 2: Pathway Analysis

```
# Load pathway analysis packages
library(pathview)
```

```
#####
Pathview is an open source software package distributed under GNU General
Public License version 3 (GPLv3). Details of GPLv3 is available at
http://www.gnu.org/licenses/gpl-3.0.html. Particullary, users are required to
formally cite the original Pathview paper (not just mention it) in publications
or products. For details, do citation("pathview") within R.
```

The pathview downloads and uses KEGG data. Non-academic uses may require a KEGG license agreement (details at <http://www.kegg.jp/kegg/legal.html>).

```
#####
```

```
library(gage)
```

```
library(gageData)
```

```
# Load KEGG pathway gene sets
data(kegg.sets.hs)
data(sigmet.idx.hs)

# Focus on signaling and metabolic pathways only
kegg.sets.hs <- kegg.sets.hs[sigmet.idx.hs]

# Examine first 3 pathways
head(kegg.sets.hs, 3)
```

```
$`hsa00232 Caffeine metabolism`
```

```
[1] "10" "1544" "1548" "1549" "1553" "7498" "9"
```

```
$`hsa00983 Drug metabolism - other enzymes`
```

```
[1] "10" "1066" "10720" "10941" "151531" "1548" "1549" "1551"
[9] "1553" "1576" "1577" "1806" "1807" "1890" "221223" "2990"
[17] "3251" "3614" "3615" "3704" "51733" "54490" "54575" "54576"
```

```
[25] "54577" "54578" "54579" "54600" "54657" "54658" "54659" "54963"
[33] "574537" "64816" "7083" "7084" "7172" "7363" "7364" "7365"
[41] "7366" "7367" "7371" "7372" "7378" "7498" "79799" "83549"
[49] "8824" "8833" "9" "978"
```

\$`hsa00230 Purine metabolism`

```
[1] "100" "10201" "10606" "10621" "10622" "10623" "107" "10714"
[9] "108" "10846" "109" "111" "11128" "11164" "112" "113"
[17] "114" "115" "122481" "122622" "124583" "132" "158" "159"
[25] "1633" "171568" "1716" "196883" "203" "204" "205" "221823"
[33] "2272" "22978" "23649" "246721" "25885" "2618" "26289" "270"
[41] "271" "27115" "272" "2766" "2977" "2982" "2983" "2984"
[49] "2986" "2987" "29922" "3000" "30833" "30834" "318" "3251"
[57] "353" "3614" "3615" "3704" "377841" "471" "4830" "4831"
[65] "4832" "4833" "4860" "4881" "4882" "4907" "50484" "50940"
[73] "51082" "51251" "51292" "5136" "5137" "5138" "5139" "5140"
[81] "5141" "5142" "5143" "5144" "5145" "5146" "5147" "5148"
[89] "5149" "5150" "5151" "5152" "5153" "5158" "5167" "5169"
[97] "51728" "5198" "5236" "5313" "5315" "53343" "54107" "5422"
[105] "5424" "5425" "5426" "5427" "5430" "5431" "5432" "5433"
[113] "5434" "5435" "5436" "5437" "5438" "5439" "5440" "5441"
[121] "5471" "548644" "55276" "5557" "5558" "55703" "55811" "55821"
[129] "5631" "5634" "56655" "56953" "56985" "57804" "58497" "6240"
[137] "6241" "64425" "646625" "654364" "661" "7498" "8382" "84172"
[145] "84265" "84284" "84618" "8622" "8654" "87178" "8833" "9060"
[153] "9061" "93034" "953" "9533" "954" "955" "956" "957"
[161] "9583" "9615"
```

Create named fold change vector for GAGE

```
# Extract log2 fold changes
foldchanges <- res$log2FoldChange

# Name fold changes with Entrez gene IDs
names(foldchanges) <- res$entrez

# Preview fold change vector
head(foldchanges)
```

```
1266      54855      1465      2034      2150      6659
```

-2.422719 3.201955 -2.313738 -1.888019 3.344508 2.392288

Run KEGG pathway analysis

```
# Run GAGE pathway analysis using KEGG gene sets
keggres <- gage(
  foldchanges,
  gsets = kegg.sets.hs
)

# Check structure of results
attributes(keggres)
```

```
$names
[1] "greater" "less"    "stats"
```

```
# View top upregulated pathways
head(keggres$greater)
```

	p.geomean	stat.mean	p.val
hsa04640 Hematopoietic cell lineage	0.002822776	2.833362	0.002822776
hsa04630 Jak-STAT signaling pathway	0.005202070	2.585673	0.005202070
hsa00140 Steroid hormone biosynthesis	0.007255099	2.526744	0.007255099
hsa04142 Lysosome	0.010107392	2.338364	0.010107392
hsa04330 Notch signaling pathway	0.018747253	2.111725	0.018747253
hsa04916 Melanogenesis	0.019399766	2.081927	0.019399766

	q.val	set.size	exp1
hsa04640 Hematopoietic cell lineage	0.3893570	55	0.002822776
hsa04630 Jak-STAT signaling pathway	0.3893570	109	0.005202070
hsa00140 Steroid hormone biosynthesis	0.3893570	31	0.007255099
hsa04142 Lysosome	0.4068225	118	0.010107392
hsa04330 Notch signaling pathway	0.4391731	46	0.018747253
hsa04916 Melanogenesis	0.4391731	90	0.019399766

```
# View top downregulated pathways
head(keggres$less)
```


	p.geomean	stat.mean	p.val
hsa04110 Cell cycle	8.995727e-06	-4.378644	8.995727e-06
hsa03030 DNA replication	9.424076e-05	-3.951803	9.424076e-05
hsa03013 RNA transport	1.375901e-03	-3.028500	1.375901e-03
hsa03440 Homologous recombination	3.066756e-03	-2.852899	3.066756e-03
hsa04114 Oocyte meiosis	3.784520e-03	-2.698128	3.784520e-03
hsa00010 Glycolysis / Gluconeogenesis	8.961413e-03	-2.405398	8.961413e-03

	q.val	set.size	exp1
hsa04110 Cell cycle	0.001448312	121	8.995727e-06
hsa03030 DNA replication	0.007586381	36	9.424076e-05
hsa03013 RNA transport	0.073840037	144	1.375901e-03
hsa03440 Homologous recombination	0.121861535	28	3.066756e-03
hsa04114 Oocyte meiosis	0.121861535	102	3.784520e-03
hsa00010 Glycolysis / Gluconeogenesis	0.212222694	53	8.961413e-03

Pathview example: Cell cycle pathway

```
# Plot KEGG cell cycle pathway
pathview(
  gene.data = foldchanges,
  pathway.id = "hsa04110"
)
```

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/joshchun/Desktop/BIMM143/class06

Info: Writing image file hsa04110.pathview.png

```
# Alternative PDF-style output
pathview(
  gene.data = foldchanges,
  pathway.id = "hsa04110",
  kegg.native = FALSE
)
```

'select()' returned 1:1 mapping between keys and columns

Warning: reconcile groups sharing member nodes!

```
      [,1] [,2]
[1,] "9"  "300"
[2,] "9"  "306"
```

Info: Working in directory /Users/joshchun/Desktop/BIMM143/class06

Info: Writing image file hsa04110.pathview.pdf

Top 5 upregulated KEGG pathways

```
# Get names of top 5 upregulated pathways
keggrespathways <- rownames(keggres$greater)[1:5]

# Extract KEGG pathway IDs
keggresids <- substr(
  keggrespathways,
  start = 1,
  stop = 8
)

# View pathway IDs
keggresids
```

```
[1] "hsa04640" "hsa04630" "hsa00140" "hsa04142" "hsa04330"
```

```
# Plot top 5 upregulated pathways
pathview(
  gene.data = foldchanges,
  pathway.id = keggresids,
  species = "hsa"
)
```

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/joshchun/Desktop/BIMM143/class06

Info: Writing image file hsa04640.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/joshchun/Desktop/BIMM143/class06

Info: Writing image file hsa04630.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/joshchun/Desktop/BIMM143/class06

Info: Writing image file hsa00140.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/joshchun/Desktop/BIMM143/class06

Info: Writing image file hsa04142.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/joshchun/Desktop/BIMM143/class06

Info: Writing image file hsa04330.pathview.png

Q7: Plot top 5 downregulated KEGG pathways

```
# Get names of top 5 downregulated pathways
down.pathways <- rownames(keggres$less)[1:5]
```

```
# Extract KEGG pathway IDs
```

```
down.ids <- substr(
  down.pathways,
  start = 1,
  stop = 8
)
```

```
# View downregulated pathway IDs
```

```
down.ids
```

```
[1] "hsa04110" "hsa03030" "hsa03013" "hsa03440" "hsa04114"
```

```
# Plot top 5 downregulated pathways
pathview(
  gene.data = foldchanges,
  pathway.id = down.ids,
  species = "hsa"
)
```

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/joshchun/Desktop/BIMM143/class06

Info: Writing image file hsa04110.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/joshchun/Desktop/BIMM143/class06

Info: Writing image file hsa03030.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/joshchun/Desktop/BIMM143/class06

Info: Writing image file hsa03013.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/joshchun/Desktop/BIMM143/class06

Info: Writing image file hsa03440.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/joshchun/Desktop/BIMM143/class06

Info: Writing image file hsa04114.pathview.png

Section 3: Gene Ontology Analysis

```

# Load GO gene sets
data(go.sets.hs)
data(go.subs.hs)

# Focus on Biological Process GO terms
gobpsets <- go.sets.hs[go.subs.hs$BP]

# Run GAGE on GO biological process terms
gobpres <- gage(
  foldchanges,
  gsets = gobpsets
)

# View top GO results
lapply(gobpres, head)

```

\$greater

	p.geomean	stat.mean	p.val
GO:0007156 homophilic cell adhesion	8.519724e-05	3.824205	8.519724e-05
GO:0002009 morphogenesis of an epithelium	1.396681e-04	3.653886	1.396681e-04
GO:0048729 tissue morphogenesis	1.432451e-04	3.643242	1.432451e-04
GO:0007610 behavior	1.925222e-04	3.565432	1.925222e-04
GO:0060562 epithelial tube morphogenesis	5.932837e-04	3.261376	5.932837e-04
GO:0035295 tube development	5.953254e-04	3.253665	5.953254e-04
	q.val	set.size	exp1
GO:0007156 homophilic cell adhesion	0.1951953	113	8.519724e-05
GO:0002009 morphogenesis of an epithelium	0.1951953	339	1.396681e-04
GO:0048729 tissue morphogenesis	0.1951953	424	1.432451e-04
GO:0007610 behavior	0.1967577	426	1.925222e-04
GO:0060562 epithelial tube morphogenesis	0.3565320	257	5.932837e-04
GO:0035295 tube development	0.3565320	391	5.953254e-04

\$less

	p.geomean	stat.mean	p.val
GO:0048285 organelle fission	1.536227e-15	-8.063910	1.536227e-15
GO:0000280 nuclear division	4.286961e-15	-7.939217	4.286961e-15
GO:0007067 mitosis	4.286961e-15	-7.939217	4.286961e-15
GO:0000087 M phase of mitotic cell cycle	1.169934e-14	-7.797496	1.169934e-14
GO:0007059 chromosome segregation	2.028624e-11	-6.878340	2.028624e-11
GO:0000236 mitotic prometaphase	1.729553e-10	-6.695966	1.729553e-10
	q.val	set.size	exp1

G0:0048285	organelle fission	5.841698e-12	376	1.536227e-15
G0:0000280	nuclear division	5.841698e-12	352	4.286961e-15
G0:0007067	mitosis	5.841698e-12	352	4.286961e-15
G0:0000087	M phase of mitotic cell cycle	1.195672e-11	362	1.169934e-14
G0:0007059	chromosome segregation	1.658603e-08	142	2.028624e-11
G0:0000236	mitotic prometaphase	1.178402e-07	84	1.729553e-10

\$stats

	stat.mean	exp1
G0:0007156 homophilic cell adhesion	3.824205	3.824205
G0:0002009 morphogenesis of an epithelium	3.653886	3.653886
G0:0048729 tissue morphogenesis	3.643242	3.643242
G0:0007610 behavior	3.565432	3.565432
G0:0060562 epithelial tube morphogenesis	3.261376	3.261376
G0:0035295 tube development	3.253665	3.253665

Section 4: Reactome Analysis

```
# Extract significant genes with adjusted p-value <= 0.05
sig_genes <- res[
  res$padj <= 0.05 & !is.na(res$padj),
  "symbol"
]

# Print total number of significant genes
print(
  paste(
    "Total number of significant genes:",
    length(sig_genes)
  )
)
```

```
[1] "Total number of significant genes: 8147"
```

```
# Save significant genes as a plain text file
write.table(
  sig_genes,
  file = "significant_genes.txt",
  row.names = FALSE,
  col.names = FALSE,
```

```
quote = FALSE  
)
```

Q. What pathway has the most significant “Entities p-value”? Do the most significant pathways listed match your previous KEGG results? What factors could cause differences between the two methods?

The pathway with the most significant “Entities p-value” in the analysis was related to the Cell Cycle/Mitotic Cell Cycle pathway. These results were consistent with the previous KEGG pathway analysis, since the Cell Cycle pathway was also one of the most significantly downregulated pathways. Other related pathways, such as DNA replication and mitosis, were also identified in both analyses. The results between Reactome and KEGG were similar overall because both analyses detected strong disruption of genes involved in cell division and cell cycle regulation after HOXA1 knockdown. However, some differences occurred because Reactome and KEGG use different pathway databases, pathway definitions, and statistical methods. Also, differences in gene annotations, pathway sizes, and significance thresholds can lead to variation in which pathways appear most enriched.